Digital Twin Benefits from Collaboration with IoT

Digital Twin is an upcoming technology that is considered to be the next big innovation. However, there is more to Digital Twin than simply providing simulation.

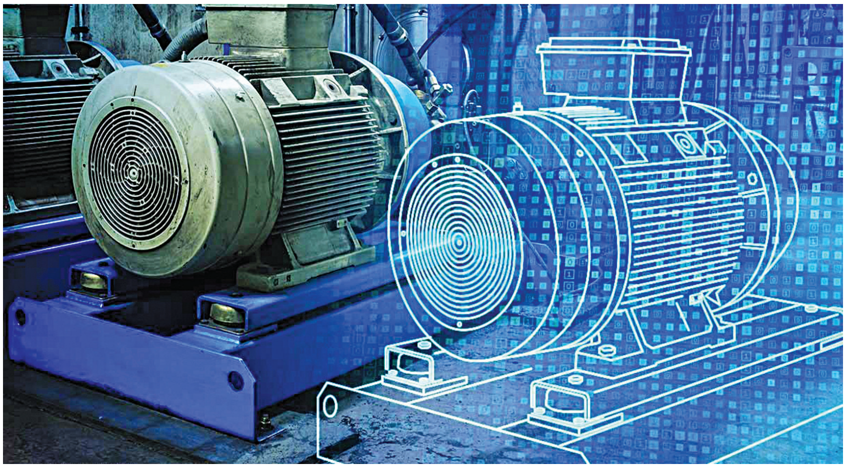

Figure 1: 3D model of a pump, which is the twin of the original (Credit: Noria Corporation)

When we hear the word 'twin,' the image of a human twin possessing identical features such as the same eyes, lips, and nose comes to our mind. Digital Twin is similar to that. It is defined as the virtual representation of a physical object or system across its entire lifecycle, which uses real-time data or data from other sources to enable learning, reasoning, and dynamic recalibration for improved decision making.

As Digital Twin is constantly evolving, its definition varies from company to company. For instance, GE states that Digital Twin is a software representation of a physical asset, system, or process designed to detect, prevent, predict, and optimise through real-time analytics to deliver business value. In simple terms, Digital Twin is defined as a virtual model of a physical thing. That thing could be a car, tunnel, bridge, jet engine, or anything else attached with multiple sensors whose function is to carry out data connection, which can be mapped to the virtual model.

But is Digital Twin all about simulation? Are physical world scenarios replicated in the digital world? Not exactly. Digital Twin initially starts as a simulation but goes on to perform real-time updates, unlike digital simulation. Results can be obtained by running, testing, and conducting assessments on a simulated version of a physical asset. Simulation is static and so does not know how the actual thing will behave in the real world. Parameters always need to be fed in the simulator to make it more realistic.

In short, a Digital Twin is the virtual representation of a system across its entire lifecycle. This includes its designing, operation, and production.

Early use of Digital Twin

In the year 1970, NASA successfully launched the spacecraft Apollo 13 into space. However, due to some technical issues, it could not complete its intended mission and had to return to Earth. It is said that NASA had used the concept of Digital Twin back then at a time when IoT did not even exist.

The story goes like this: Before the space mission began, a simulator was built to train the chosen astronauts to face any emergency. On the day of the big event, the astronauts aboard Apollo 13 had access to only its telemetry data. Fortunately, the same telemetry data was accessible by the ground station on the Earth as well. Despite being far apart by several thousand kilometres, communication was established wherein exchange of messages took place thanks to virtual representation. Before giving various instructions, verification was done. If the results were satisfactory, only then communication ensued. This constant contact played a big role in bringing the Apollo 13 spacecraft safely back to Earth.

Since there was no use of IoT, it could not technically be called a Digital Twin. But it qualifies as an example of Digital Twin as it combined multiple things for communication purposes. This made the whole system capable of multitasking and responsive.

In 1977, several flight simulators were introduced, which allowed pilots to practice their flight techniques. In 1982, AutoCAD was launched, which eased the design of 2D and 3D models in the times to come.

In 2002, Dr Grieves introduced the first concept of Digital Twin and in 2011, NASA published several papers on the subject. In 2015, GE became the first company to take the Digital Twin initiative. After 2015, several industrial organisations started adopting Digital Twin as part of their operations.